Evaluating Methods for Question Answering over Large Code Repositories and Technical Documents

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<https://github.com/muoimeo/evaluating-long-context-rag-and-recursive-reading-for-llm-repository-qa.git>

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Abstract

This project investigates methods for question answering over large code repositories and technical documents, with a focus on maintaining grounded and reliable responses as context size increases. Three approaches are evaluated: (1) a naive long-context baseline that processes large inputs in a single pass, (2) a standard Retrieval-Augmented Generation (RAG) pipeline that retrieves relevant chunks before answering, and (3) a bounded structure-aware iterative reading method that performs targeted evidence expansion using deterministic repository navigation tools.

The central research problem is “context rot”, where model performance degrades as input length grows and distractor content increases. To study this, the project constructs a benchmark of repository and document-level questions and evaluates each method on answer correctness, citation quality, latency, and model usage. Citation evaluation is designed to account for chunk-level retrieval by using containment-based metrics.

Results show that while RAG provides a strong and efficient baseline, the iterative reading method offers advantages on multi-step and trace-style questions by improving evidence coverage. However, these gains come at increased computational cost. The findings highlight trade-offs between accuracy, grounding, and efficiency, and suggest that structured iterative reading is beneficial when retrieval alone is insufficient.

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Introduction

Modern software repositories and technical documents have grown to a scale where manual navigation is increasingly impractical. Developers working with unfamiliar codebases often need to answer questions such as “Where is authentication enforced?” or “How are retries implemented?”. Similarly, long technical documents require significant effort to locate precise supporting information. While large language models (LLMs) provide a natural interface for such queries, their effectiveness is constrained by context limitations and their sensitivity to long or noisy inputs.

A key challenge in applying LLMs to this setting is not only generating correct answers, but ensuring that answers are grounded in verifiable evidence. This becomes more difficult as the size of the corpus increases. As more text is included, models are more likely to overlook relevant information, attend to irrelevant content, or produce unsupported claims. This phenomenon, often referred to as “context rot”, motivates the need for methods that can scale beyond naive long-context prompting.

This project investigates three approaches to question answering over large repositories and technical documents. The first is a single-pass long-context baseline, where large portions of the corpus are directly included in the prompt. This approach is simple but suffers from known degradation as input length increases. The second is a standard Retrieval-Augmented Generation (RAG) pipeline, which retrieves relevant chunks and conditions the model on these passages, improving grounding and scalability. The third approach is a bounded structure-aware iterative reading method. Instead of relying on a single retrieval step, this method performs a limited sequence of evidence expansion steps guided by candidate symbols such as functions, classes, or file paths. Deterministic repository navigation tools are used to locate definitions and references before synthesizing a final answer.

Importantly, the proposed method is not a fully autonomous recursive agent. It is a practical and bounded approximation designed for local inference, focusing on structured evidence expansion rather than unconstrained multi-step reasoning. The goal is to evaluate whether this additional structure provides measurable benefits over standard approaches.

The primary research question is whether iterative, structure-aware reading improves answer correctness and citation quality compared to long-context prompting and RAG, particularly as context complexity increases. Secondary questions examine how these methods degrade under increasing corpus size and distractor content, and what trade-offs arise in latency and computational cost.

To answer these questions, the project constructs a benchmark of repository and document-based questions, including both single-hop and multi-hop cases. Each method is evaluated on answer correctness, citation quality, latency, and model usage. By comparing performance across

controlled conditions, the project aims to provide practical insights into when iterative reading is beneficial and when simpler approaches remain sufficient.

Literature Review

The contradiction: Long-Context models vs. Standard RAG

Recent advancements in Large Language Models (LLMs) have substantially expanded context windows, theoretically enabling the ingestion of entire code repositories into a single prompt. However, empirical evaluations reveal a fundamental contradiction: merely expanding context does not equate to improved reasoning. Research on the "lost in the middle" phenomenon demonstrates that LLM performance degrades significantly when critical evidence is buried within long, noisy contexts, a vulnerability often termed "context rot" (Liu et al., 2024; Du et al., 2025). Conversely, standard Retrieval-Augmented Generation (RAG) mitigates context dilution by isolating relevant chunks via vector search (Lewis et al., 2020). Yet, standard RAG introduces its own brittleness in software engineering tasks. Studies show that continuously adding retrieved passages yields diminishing returns, as chunk-based retrieval often fragments logical dependencies and surfaces semantically similar but logically disconnected "hard negatives" (Jin et al., 2024; Li et al., 2024). This contradiction highlights that neither raw long-context prompting nor static one-shot RAG is sufficient for complex, multi-file codebase comprehension. Therefore, establishing these two paradigms as baselines (Method A and Method B) is crucial to empirically quantify their failure modes under controlled distractor noise.

The evolution: Toward Structure-Aware Iterative Reading

To overcome the limitations of static retrieval and long-context degradation, the field is evolving toward interleaved reading, reasoning, and retrieval (Trivedi et al., 2023). Frameworks such as ReAct (Yao et al., 2023) and Toolformer (Schick et al., 2023) pioneered the use of LLMs as agents capable of executing targeted tool calls. Building on this, adaptive mechanisms like Adaptive-RAG dynamically adjust retrieval strategies based on query complexity (Jeong et al., 2024), while GraphReader demonstrates the necessity of maintaining a structured, evidence-bound state to navigate long-context dependencies selectively (Li et al., 2024). Furthermore, recent findings by Qiu et al. (2025) on In-Context Retrieval and Reasoning (ICR^2) emphasize that allowing models to actively filter and de-noise contexts during generation significantly outperforms naive RAG. However, fully autonomous recursive agents are computationally prohibitive and prone to cascading reasoning errors under local inference constraints. Consequently, this project implements Method C: a bounded, structure-aware iterative reader that leverages deterministic code-navigation tools to achieve the robustness of agentic reading without the prohibitive costs of unbounded recursion.

The evaluation gap: Factual grounding over N-Grams

A major unresolved gap in repository QA lies in the evaluation methodology. Standard n-gram metrics (BLEU, ROUGE) and semantic overlap metrics (BERTScore) correlate poorly with factual correctness in highly structured domains like code. Evaluating these systems requires a paradigm shift towards claim-level attribution. The ALCE benchmark established the necessity of evaluating fluency, correctness, and citation quality as distinct dimensions, showing that even state-of-the-art LLMs frequently hallucinate unsupported claims (Gao et al., 2023). More recently, fine-grained diagnostic frameworks like RAGChecker (Ru et al., 2024) and ARES (Saad-Falcon et al., 2024) have standardized the use of composite metrics to separate retrieval omissions from synthesis hallucinations. Without explicit citation metrics, an answer might appear semantically plausible yet completely lack factual grounding. To address this gap, our evaluation pipeline adopts an LLM-as-a-judge for semantic correctness alongside deterministic geometric metrics (File F1 and Span containment) inspired by RAGChecker and ALCE, explicitly quantifying how accurately each method grounds its claims in the source repository.

Methodology

System Architecture and Data Ingestion

To empirically evaluate the robustness of question-answering strategies over large technical corpora, a unified end-to-end pipeline was constructed. The dataset comprises an open-source code repository (FastAPI) and a comprehensive technical documentation set (AWS Lambda Developer Guide). The data ingestion phase parses supported files while excluding environment-specific noise (e.g., node_modules, .git). Documents are segmented into chunks using an overlapping sliding window, with each chunk strictly retaining its absolute file path and line-number metadata (line_start, line_end). The chunks are embedded using the BAAI/bge-small-en-v1.5 model and indexed within a persistent ChromaDB vector store.

Evaluated Question Answering Strategies

To ensure a fair and rigorous comparison, Method B and Method C share the identical foundational retrieval pipeline. They utilize the same chunker, the same embedding model, the same base dense retriever, the same heuristic reranker (which applies calibrated penalties to low-signal files and bonuses to core source code), and the exact same initial candidate pool size. This shared foundation guarantees that any performance divergence between the methods stems entirely from their reading and reasoning architectures, not from an unequal baseline access to the corpus.

Method A: Long-Context Prompting Baseline

Method A evaluates the capacity of LLMs to natively synthesize information from massive contexts without explicit retrieval. Bound by a strict 50,000-token budget, the system employs a "round-robin" context assembly strategy to shuffle and interleave chunks from the available corpora deterministically. Instead of a summarize-then-answer approach, the system utilizes direct context-stuffing. The LLM is prompted to read this monolithic context and output a structured JSON response in a single pass, adhering to a predefined schema containing both the final answer and its corresponding evidence citations. If structured parsing fails due to context-length degradation, the system gracefully falls back to an answer-only generation mode.

Method B: Standard Retrieval-Augmented Generation (RAG)

Method B represents the industry-standard single-pass retrieval approach. After retrieving the initial candidate pool and applying the shared reranker, it strictly truncates the context to a minimal, high-value subset (e.g., maximum 4 chunks). The prompt is dynamically adjusted based on a

query classifier (extract, trace, compare). The LLM synthesizes the final answer in a single generation step, restricted strictly to the provided chunks. This method serves as a highly efficient, low-latency baseline.

Method C: Structure-Aware Iterative Reader

Rather than a fully autonomous agent, Method C is a deterministic, state-driven iterative reading pipeline. Starting from the same initial candidate pool as Method B, Method C evaluates each chunk individually, classifying it as "direct" evidence, "bridge" evidence, or "noise". It extracts typed facts and maintains an evidence-bound search agenda.

To resolve missing relations, Method C dynamically expands its context through follow-up retrievals. While the system does incorporate deterministic code-navigation tools (e.g., find_definition, read_neighbors), the relative contribution of these tools versus iterative follow-up retrieval is examined separately in the ablation study. Method C's strength lies in its ability to selectively accumulate verified facts across multiple steps before synthesizing the final answer, avoiding the pitfalls of single-pass generation. Crucially, this bounded two-stage design directly addresses vulnerabilities identified in recent literature; it captures the noise-filtering benefits of in-context retrieval and reasoning (Qiu et al., 2025) while explicitly preventing the cascading reasoning errors typical of fully unconstrained autonomous agents (Trivedi et al., 2023; Yao et al., 2023).

Evaluation Framework and Metrics

The performance of the three methods is quantified using an automated, multi-dimensional evaluation harness (score.py) that separates answer correctness from citation accuracy.

For Answer Correctness, the pipeline utilizes the LLM-as-a-Judge paradigm (Zheng et al., 2023). The judge evaluates the predicted answer against a ground-truth reference and grading notes using a strict rubric: 1.0 (fully correct, capturing the core mechanism without major factual errors), 0.5 (partially correct, capturing relevant ideas but missing required implementation details or tracing steps), or 0.0 (incorrect, unsupported, or answering at the wrong abstraction level). To ensure reliability, a human subset evaluation is conducted to compute inter-rater agreement (e.g., Cohen's Kappa) between the LLM judge and human experts. Furthermore, a consistency check and fallback mechanism is implemented: if the primary judge model encounters a transient error or output parsing failure, the system automatically falls back to a secondary model, explicitly logging the judge's status to maintain evaluation integrity.

For Citation Quality, addressing the factual grounding evaluation gap highlighted by recent literature (Yen et al., 2025), the system moves beyond inadequate lexical metrics like BLEU or ROUGE. Instead, it implements a suite of geometric, deterministic citation metrics

(verify_evidence.py) to strictly evaluate claim-level attribution. It calculates document-level precision, recall, and F1, analogous to the retrieval diagnostics in modern frameworks (Ru et al., 2024), along with span-level containment and F2-style overlap metrics based on exact file paths and line-number overlaps between predicted and reference evidence.

Furthermore, a weighted span score is computed to effectively balance containment-based behavior and precision, directly implementing the rigorous citation fidelity principles established by contemporary benchmarks (Gao et al., 2023; Saad-Falcon et al., 2024). This ensures that models are not unfairly rewarded for citing excessively large blocks of code.

Context-Rot Experimental Design

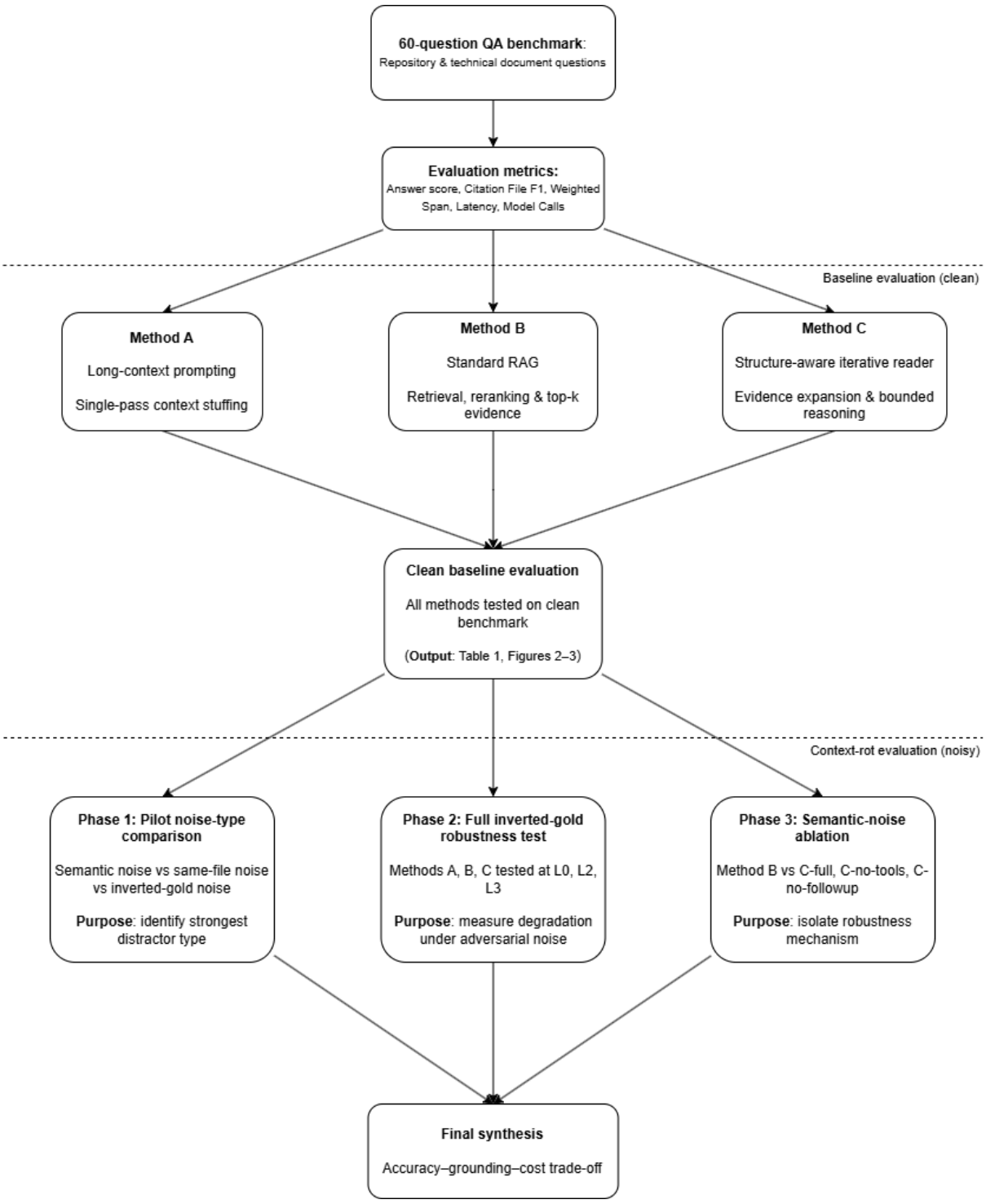


Figure 1*: Experimental design linking the three evaluated methods to the clean baseline, context-rot phases, and final robustness analysis.*

To isolate the "context rot" phenomenon and evaluate adversarial robustness, a controlled context-perturbation experiment is executed. Instead of using random, irrelevant text, the experiment injects structured distractors reflecting real-world repository challenges. The evaluation is systematically staged across three distinct phases:

Phase 1: Pilot Noise-Type Comparison.

A pilot study on a subset of questions evaluating different noise paradigms: *semantic noise* (chunks similar to the query but lacking the answer), *same-file noise* structurally adjacent chunks that may act as hard negatives, although in some cases they also provide local bridging context), and *inverted-gold noise* (adversarial chunks mirroring gold evidence but with inverted logic). This phase identifies the most informative failure modes.

Phase 2: Full-Scale Inverted-Gold Evaluation.

The primary robustness test running all 60 benchmark questions across Methods A, B, and C. The evaluation is staged at L0 (clean core, overlapping gold evidence only), L2, and L3 (heavy injection of inverted-gold distractors). By restricting Methods B and C to retrieve strictly from this bounded, noisy pool, this phase empirically measures performance degradation under adversarial conditions.

Phase 3: Semantic Noise Ablation.

A targeted 60-question ablation study at L0 and L3 comparing Method B with variants of Method C (e.g., *C-full*, *C-no-tools*, *C-no-followup*). This phase investigates the underlying mechanisms of robustness, exploring whether the deterministic tools or the iterative follow-up retrieval processes contribute more significantly to the system's resilience against semantic distractors.

Results and Analysis

This section evaluates the three question-answering strategies under both standard conditions and adversarial context perturbations. The analysis proceeds through four stages: establishing baseline performance, profiling distractor types, evaluating robustness under context rot, and an ablation study to isolate the mechanisms of Method C. The objective is to assess not only answer correctness but also citation precision and computational efficiency.

Standard Benchmark Results

The initial evaluation was conducted on a standard 60-question benchmark with clean contexts. Results demonstrate that the Structure-Aware Iterative Reader (Method C) achieved the strongest overall performance, yielding an answer score of 0.842, compared to 0.783 for Standard RAG (Method B) and 0.425 for the Long-Context baseline (Method A). Citation grounding followed a similar trajectory: Method C delivered superior evidence localization with a Document-level F1 (File F1) of 0.737 and a Weighted Span score of 0.597, outperforming Method B (0.530 and 0.484, respectively). Method A demonstrated the weakest span grounding (0.131), struggling to pinpoint exact evidence despite having access to the entire context.

Method	Answer Score	Citation File F1	Weighted Span Score	Latency	Model Calls
Long-context Baseline (A)	0.425	0.317	0.131	35.0s	**1.0**
Standard RAG (B)	0.783	0.530	0.484	**12.1s**	**1.0**
Structure-Aware Iterative Reader (C)	**0.842**	**0.737**	**0.597**	93.6s	11.1

Table 1: *Standard 60-question benchmark results comparing the long-context baseline, standard RAG, and the structure-aware iterative reader.*

However, Method C's quality gains incur a severe computational premium. Method B remains the most practical baseline, operating with a mean latency of 12.1 seconds and exactly 1.0 model call per question. Conversely, Method C requires an average of 11.1 sequential model calls, escalating latency to 93.6 seconds. Method A was slower than B (35.0 seconds) yet still produced the weakest answers. Consequently, while Method C performed best for factual precision in this benchmark, Method B remains superior in efficiency. (*Note: BERTScore remained relatively high across all methods, proving insensitive to implementation-level factual errors; it is therefore treated only as a secondary sanity-check metric*)

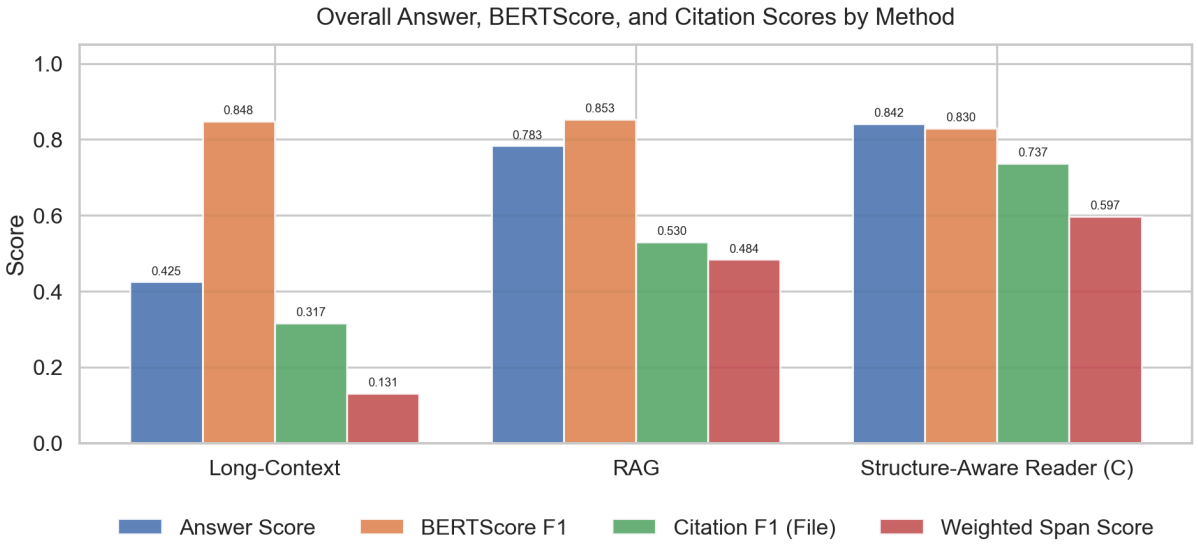


Figure 2: *Overall benchmark comparison of answer correctness, BERTScore, and citation metrics across the three methods.*

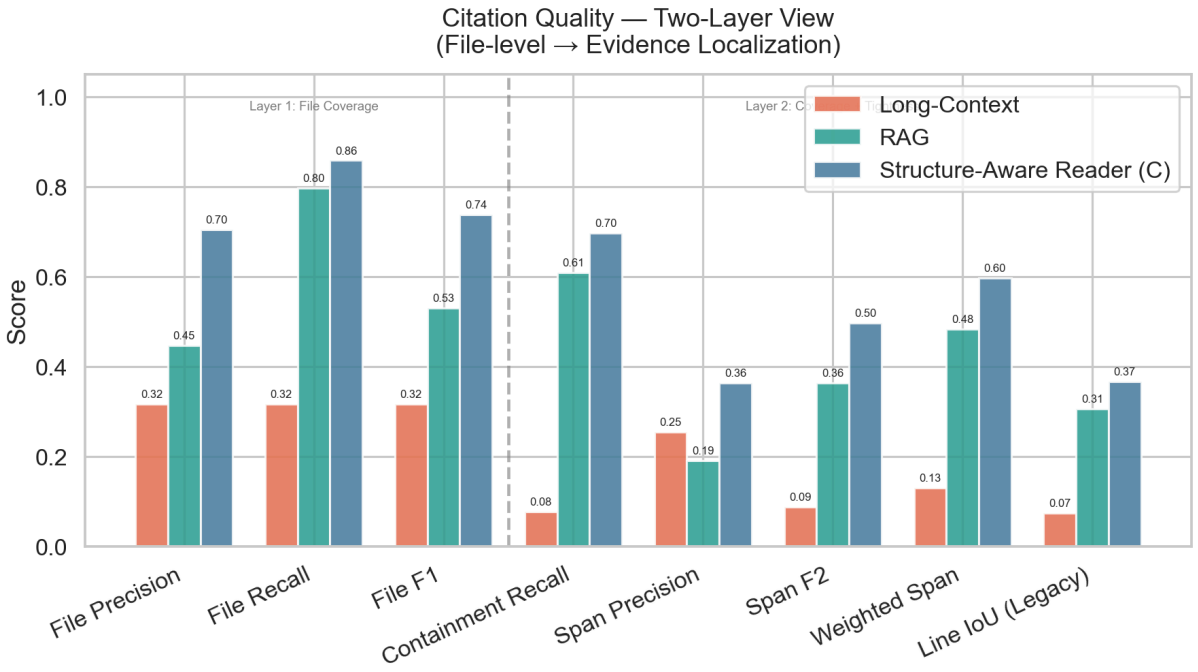


Figure 3*: Two-layer view of citation quality, separating file-level evidence retrieval from span-level evidence localisation.*

Context-rot Phase 1: Profiling Distractor Types

A 12-question pilot study investigated three noise paradigms. Under semantic noise, Method B's answer score degraded from 0.875 to 0.667 at Level 3 (L3), whereas Method C remained entirely stable (0.750 to 0.750). Interestingly, same-file noise acted less as a distractor and more as local context expansion, inadvertently improving Method C's answer score (0.750 to 0.833) by providing useful bridging logic.

Inverted-gold noise (adversarial chunks mirroring genuine evidence but reversing its logic) emerged as the most disruptive condition. It collapsed Method B's Weighted Span score from 0.584 to 0.075 and Method C's from 0.688 to 0.250. Because it effectively poisoned the synthesis stage, inverted-gold noise was selected as the primary adversarial stress test for Phase 2.

Context-rot Phase 2: Adversarial Robustness under Context Rot

The full 60-question robustness experiment evaluated the methods under escalating inverted-gold noise (L0 to L3). By isolating retrieval from a shared, noisy candidate pool, the experiment directly tested synthesis resilience.

Under clean conditions (L0), Method B achieved the highest answer score (0.900), followed closely by C (0.858). However, under heavy adversarial noise (L3), Method B's performance plummeted to 0.642, while Method C retained a higher score of 0.692. The contrast is starkest at the evidence level: Method B's File F1 collapsed from 0.698 to 0.082, and its Weighted Span dropped from 0.670 to 0.091. Method C degraded much more gracefully, retaining a File F1 of 0.255 and a Weighted Span of 0.234.

Method A presented an "illusion of robustness" - its answer score declined only slightly (0.825 to 0.775), and its File F1 remained high (0.946 to 0.932). However, this merely indicates the correct files were physically present in the massive prompt; its Weighted Span dropped severely (0.574 to 0.284), proving it failed to accurately localize evidence. Ultimately, Method C is significantly more robust than one-shot RAG because its selective reading prevents adversarial chunks from seamlessly poisoning the final answer generation.

Method	L0 Answer ↑	L3 Answer ↑	Δ Answer	L0 File F1 ↑	L3 File F1 ↑	Δ File F1	L0 Weighted Span ↑	L3 Weighted Span ↑	Δ Span
Long-context Baseline (A)	0.825	**0.775**	**-0.050**	**0.946**	**0.932**	**-0.014**	0.574	**0.284**	**-0.290**
Standard RAG (B)	**0.900**	0.642	-0.258	0.698	0.082	-0.616	0.670	0.091	-0.579
Structure-Aware Iterative Reader (C)	0.858	0.692	-0.167	0.859	0.255	-0.604	**0.712**	0.234	-0.478

Table 2*: Performance degradation from L0 to L3 under full inverted-gold context-rot evaluation. Negative Δ values indicate degradation under adversarial noise.*

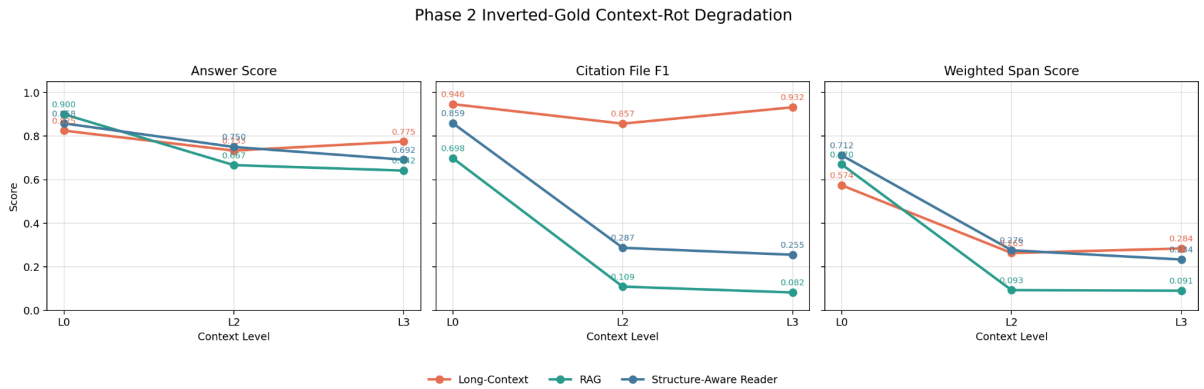


Figure 4: *Context-rot degradation curves for Methods A, B, and C under inverted-gold noise across L0, L2, and L3.*

Context-rot Phase 3: Semantic Noise Ablation

To isolate the mechanisms driving Method C's robustness, an ablation study was conducted under full 60-question semantic noise. Method C was decoupled into *C-full*, *C-no-tools* (removing deterministic code navigation), and *C-no-followup* (removing iterative evidence expansion).

At L3, *C-full* consistently outperformed Method B in answer score (0.867 vs 0.800) and Weighted Span (0.682 vs 0.546). Crucially, the *C-no-tools* variant matched C-full exactly in answer score (0.867) and even slightly exceeded it in citation metrics. By contrast, *C-no-followup* suffered a notable drop in grounding (Weighted Span fell to 0.635). These empirical results confirm that deterministic lexical tools are not the primary driver of robustness. The true causal mechanism protecting the system is the iterative follow-up and evidence-centric synthesis - the ability to maintain an agenda of unresolved claims and selectively accumulate facts over multiple steps.

Method	L0 Answer ↑	L3 Answer ↑	Δ Answer	L0 File F1 ↑	L3 File F1 ↑	Δ File F1	L0 Weighted Span ↑	L3 Weighted Span ↑	Δ Span
B RAG	0.892	0.800	-0.092	0.713	0.560	-0.153	0.678	0.546	-0.132
C full	0.850	**0.867**	**+0.017**	0.864	0.824	-0.040	0.714	0.682	-0.032
C no-tools	0.842	**0.867**	**+0.025**	0.864	**0.832**	**-0.031**	0.714	**0.696**	**-0.018**
C no-followup	0.858	0.850	-0.008	**0.880**	0.800	-0.080	**0.725**	0.635	-0.090

Table 3*: Semantic-noise ablation at L0 and L3 comparing Method B with the full and ablated variants of Method C.*

Failure Taxonomy and Synthesis

A post-hoc heuristic failure taxonomy illuminates these operational differences. Under Phase 2 inverted-gold noise (L3), 48 out of 60 runs for Method B were classified as distractor_selection_failure, compared to only 30 for Method C. This empirically demonstrates that static top-k retrieval is highly brittle when hard negatives closely match the semantic profile of gold evidence.

Under the milder Phase 3 semantic noise, Method B exhibited 7 distractor failures and 6 evidence localization failures. Method C-full and C-no-tools were remarkably stable, logging 54 and 55 successful runs respectively, with merely 3 distractor failures. However, C-no-followup recorded only 48 successes, with a spike in evidence-localisation failures.

In conclusion, while Method B remains the optimal strategy for latency-sensitive applications in clean environments, it is fundamentally vulnerable to context poisoning. Method C successfully mitigates this vulnerability - not through complex tool calling, but through iterative evidence expansion and strict state-bounding - albeit at a significant computational cost.

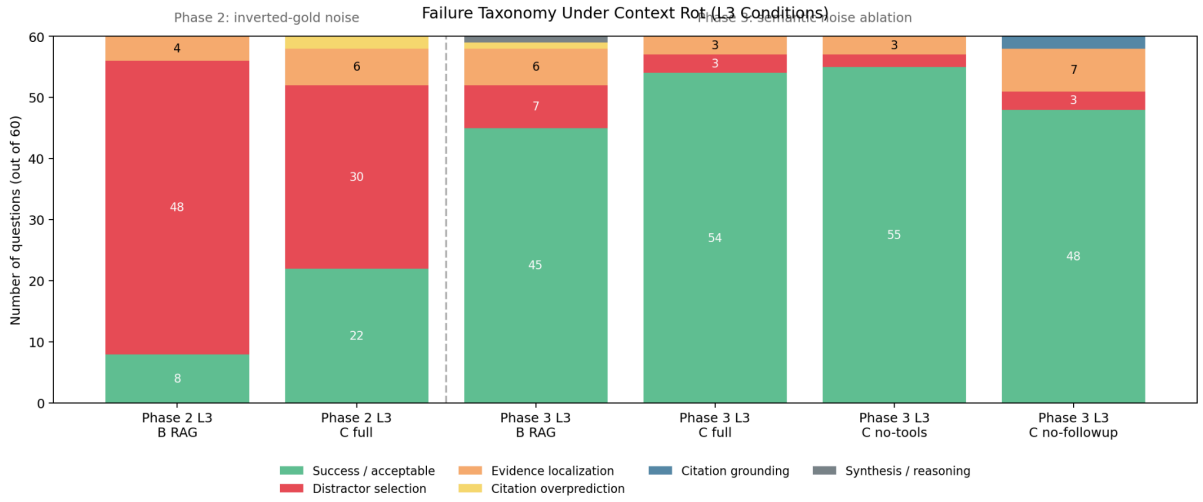


Figure 5*: Heuristic post-hoc failure taxonomy under L3 conditions for Phase 2 and Phase 3.*

Project Evaluation

A central strength of this project is that evaluation was treated as an evolving, reflexive part of the research design rather than a final reporting step. As reflected in the methodology and results sections, the system architecture, benchmark datasets, and evaluation protocols were repeatedly revised in response to intermediate empirical findings.

One major refinement was the architectural shift from an open-ended recursive reading concept to the bounded, structure-aware iterative reader deployed as Method C. This was not merely a practical compromise, but a methodological necessity: early testing revealed that unbounded recursion was difficult to control, computationally prohibitive under local inference, and prone to cascading reasoning errors, a limitation well-documented in multi-step reasoning and agentic frameworks (Trivedi et al., 2023; Yao et al., 2023). The final bounded design preserved the core advantage of multi-step evidence expansion while making the method deterministic, computationally budgeted, and suitable for rigorous systematic comparison against Methods A and B.

The project also actively adapted to infrastructure constraints in a manner that ultimately improved evaluation quality. Early dependence on external commercial API models introduced rate limits, variable latency, and limited reproducibility. Migrating the core pipeline to a local model setup significantly improved experimental control. It enabled repeated scoring and robust adversarial testing, although it concurrently forced stricter operational limits on model calls and follow-up retrieval depth, directly shaping the efficient design of Method C.

The evaluation protocol itself underwent rigorous refinement. Early iterations revealed issues with stale artifacts, inconsistent JSON schemas, and weak evidence spans, highlighting that poor preliminary results often reflected pipeline defects rather than actual model behaviour. In response, the evaluation harness was tightly standardised, benchmark evidence was manually verified, and consistency checks with fallback mechanisms were introduced. This reflexive debugging crucially safeguarded the validity of the final comparative analyses.

Furthermore, metric selection was treated with critical scrutiny. Traditional n-gram metrics (BLEU, ROUGE) and semantic overlap metrics (BERTScore) were initially considered but ultimately rejected as primary indicators. Empirical observation - supported by recent benchmarking studies in repository and long-context QA (Yen et al., 2025) - demonstrated that lexical metrics correlate poorly with factual correctness, and BERTScore remained artificially high even when implementation logic or evidence grounding failed entirely. Consequently, the final evaluation prioritised rubric-based LLM-as-a-judge scoring alongside deterministic citation metrics at both the file and span levels (Gao et al., 2023), retaining BERTScore merely as a secondary sanity-check.

Finally, the phase-based context-rot experimental design illustrates iterative reflective practice. Phase 1 diagnostics revealed that "same-file noise" did not function as a pure hard-negative

condition, while "inverted-gold noise" provided the most informative adversarial stress test. This insight directly motivated the full-scale Phase 2 robustness evaluation. Phase 3 then subjected Method C to a critical ablation study, which purposefully prevented overclaiming: it proved that removing deterministic tools had minimal effect, whereas disabling iterative follow-up retrieval severely weakened grounding (Ru et al., 2024). Overall, the project demonstrates a continuous cycle of reflection, correction, and redesign, ensuring the final conclusions are the product of a rigorous process aimed at maximizing fairness, validity, and explanatory depth.

Further Work and Conclusion

This project evaluated long-context prompting, standard RAG (Method B), and a bounded iterative reader (Method C) for repository question answering. On the standard benchmark, Method C achieved the highest answer quality and citation grounding. Under adversarial context-rot conditions - especially inverted-gold noise - Method B degraded sharply, whereas Method C demonstrated graceful degradation. However, Method C incurs a substantial computational premium. Thus, bounded iterative reading appears to be an effective approach for improving grounding robustness, though it does not entirely eliminate adversarial failure modes.

Several directions for future research emerge directly from these findings. First, Method C remains vulnerable to "blind trust" in semantically relevant but factually incorrect distractors (Ru et al., 2024). Future systems should integrate explicit, claim-level contradiction detection before synthesis, utilizing diagnostic frameworks like RAGChecker (Ru et al., 2024) to ensure strict citation alignment. Second, Method C's latency is a major operational bottleneck. Future work must explore adaptive routing mechanisms - such as Adaptive-RAG (Jeong et al., 2024) or Self-ROUTE (Li et al., 2024) - to dynamically direct simple queries to single-pass RAG and reserve computationally expensive iterative reading exclusively for complex, multi-hop reasoning.

Third, the evaluation framework requires further maturation. Future iterations should scale the benchmark to match comprehensive development datasets like SWE-bench (Jimenez et al., 2024), and meticulously calibrate LLM-as-a-judge scoring via human-preference frameworks like ARES (Saad-Falcon et al., 2024) to ensure high inter-rater reliability. Additional ablations isolating dense retrieval from ColBERT-style late-interaction reranking (Santhanam et al., 2022) would further clarify performance drivers. Ultimately, structured iterative reading is highly promising for noisy repository QA, but it is best conceptualized as a premium robustness strategy rather than a blanket replacement for standard RAG.

Glossary

Term	Definition
Context rot	Degradation in model performance as input context becomes longer, noisier, or more distractor-heavy.
Grounding	The extent to which a generated answer is supported by verifiable source evidence.
Retrieval-Augmented Generation (RAG)	A QA pipeline that retrieves relevant chunks from a corpus before generating an answer.
Long-context baseline	A baseline method that places a large amount of context into a single prompt rather than retrieving targeted evidence per query. In this project, it corresponds to Method A.
Chunk	A small unit of text or code produced during ingestion. Each chunk retains metadata such as file path, line range, chunk index, and content.
Top-k retrieval	A retrieval mechanism that selects the k most relevant chunks for a query.
Hard negative	A retrieved item that is semantically similar to the query or gold evidence but does not actually support the correct answer.
Inverted-gold noise	Adversarial distractor evidence that resembles gold evidence while reversing or contradicting its logic.
Semantic noise	Irrelevant or partially relevant distractor chunks that are topically similar to the query but do not contain the correct answer.
Single-hop question	A question answerable from one primary evidence location.
Multi-hop question	A question requiring evidence integration from multiple files, sections, or reasoning steps.
Gold evidence	The benchmark reference evidence, defined as the file paths and line spans needed to support the ground-truth answer.
Evidence localisation	The ability of a system to cite not only the right file but the correct supporting span within that file.
LLM-as-judge	An evaluation approach in which a language model assesses the semantic correctness of a predicted answer against the ground truth, rather than relying purely on lexical overlap.
Ablation study	An experiment that removes or disables components of a system to determine which parts are causally responsible for performance.

Table of Abbreviations

Abbreviation	Meaning
LLM	Large Language Model
RAG	Retrieval-Augmented Generation
QA	Question Answering
F1	Harmonic mean of precision and recall
F2	F-beta score with beta = 2
IoU	Intersection over Union
GT	Ground Truth
AWS	Amazon Web Services
ALCE	Automatic LLMs' Citation Evaluation
CRAG	Corrective Retrieval-Augmented Generation
BERTScore	Bidirectional Encoder Representations from Transformers Score
L0 / L1 / L2 / L3	Experimental noise levels used in context-rot evaluation

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Appendix A: Supplementary Figures and Diagnostic Plots

This appendix provides supplementary diagnostic plots that support the main findings without interrupting the main results section. These figures provide additional evidence about latency distribution, dataset-specific citation behaviour, and method-level error patterns.

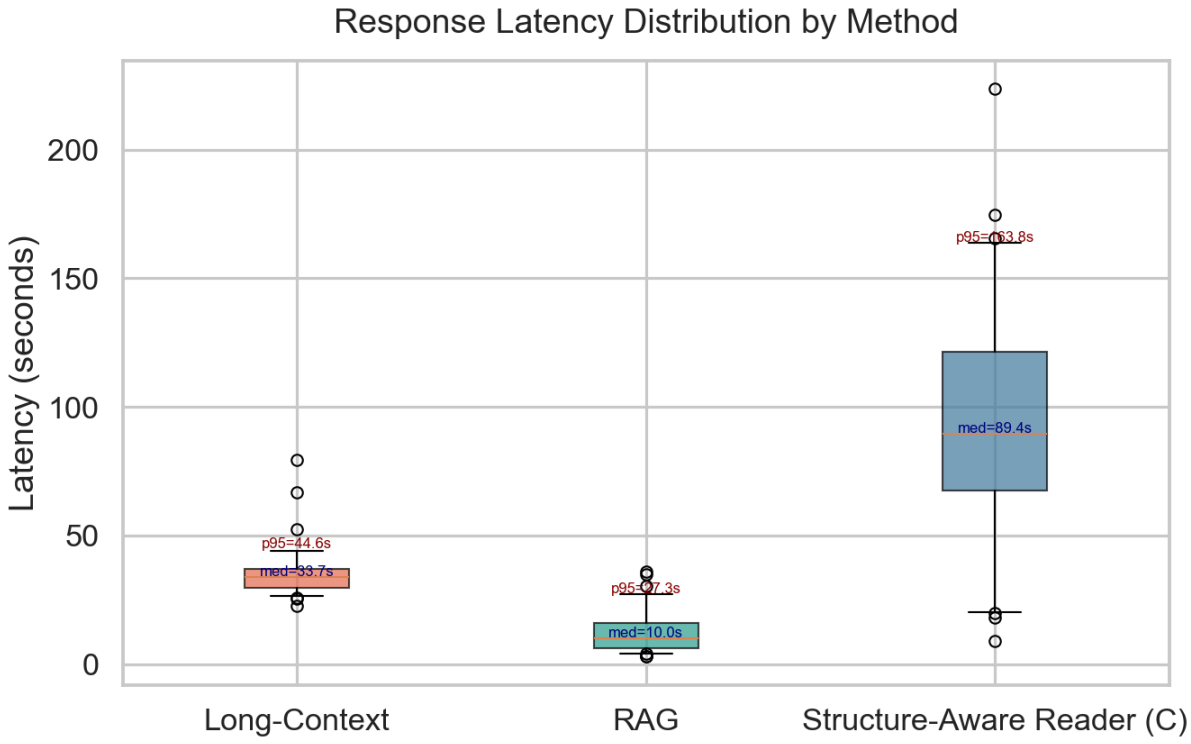


Figure A1: *Response latency distribution by method.*

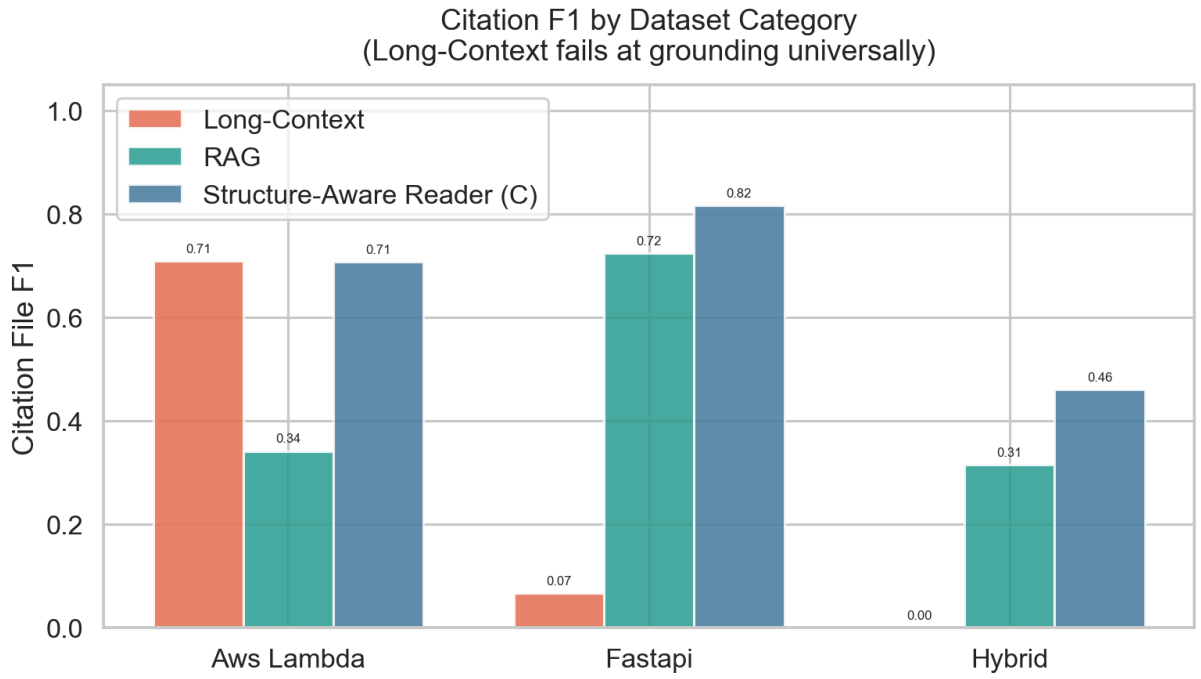


Figure A2: *Citation File F1 by dataset category.*

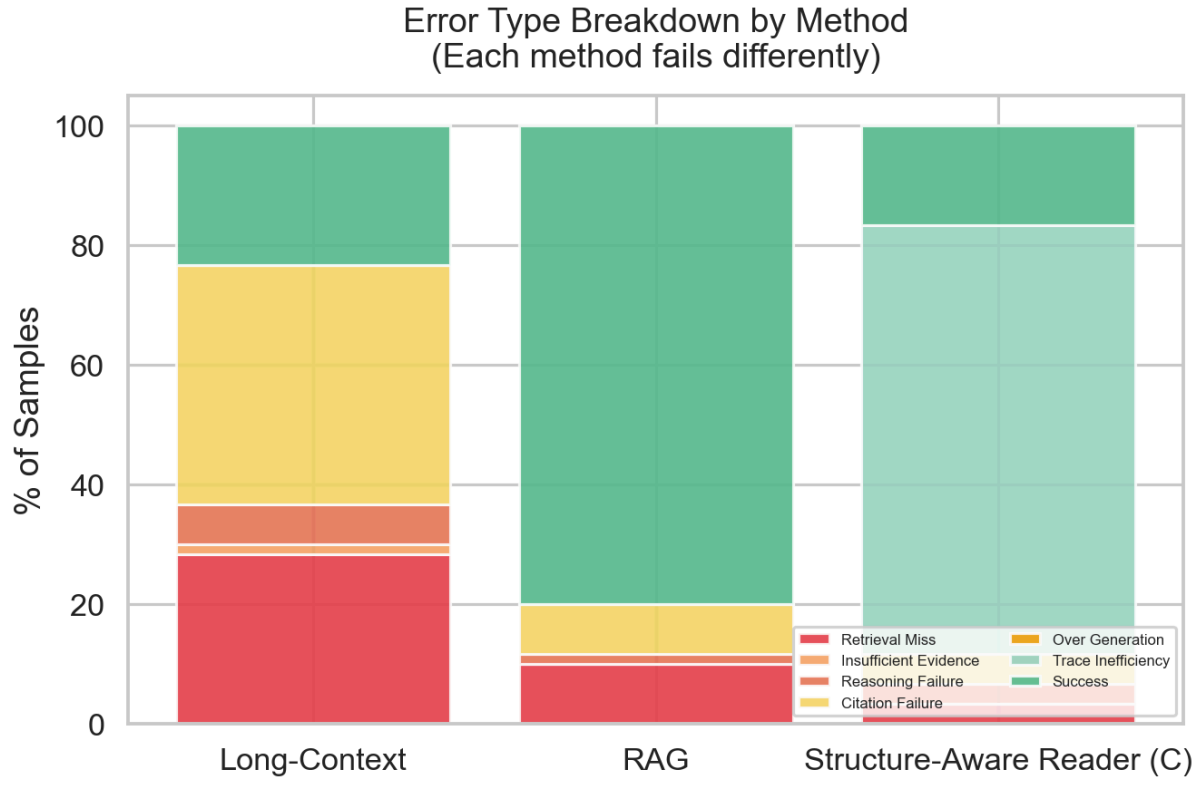


Figure A3: *Error type breakdown by method.*

Appendix B: Metric Definitions and Evaluation Protocol

Metric	Level	Purpose
Answer score	Answer	Measures semantic correctness of the predicted answer using an LLM-judge rubric scored as 0 / 0.5 / 1.
Citation File F1	File	Measures whether the method cites the correct source files.
Containment Recall	Span	Measures whether predicted citation spans contain the required gold evidence lines.
Span Precision	Span	Measures whether predicted citation spans avoid excessive irrelevant lines.
Weighted Span Score	Span	Main span-level grounding metric; balances evidence coverage and citation tightness with higher weight on recall.
Latency	System	Measures end-to-end response time per question.
Model Calls	System	Measures computational cost through the number of LLM calls.

Table B1: *Summary of evaluation metrics.*

Appendix C: Extended Context-Rot Results

Method	Level	Answer Score	Citation File F1	Weighted Span Score	Latency (s)	Model Calls
Long-context Baseline (A)	L0	0.825	**0.946**	0.574	15.486	1.000
	L2	0.733	**0.857**	0.263	33.006	1.000
	L3	**0.775**	**0.932**	**0.284**	31.980	1.000
Standard RAG (B)	L0	**0.900**	0.698	0.670	**14.453**	1.000
	L2	0.667	0.109	0.093	**14.214**	1.000
	L3	0.642	0.082	0.091	**12.981**	1.000
Structure-aware Iterative Reader (C)	L0	0.858	0.859	**0.712**	72.671	8.867
	L2	**0.750**	0.287	**0.276**	121.300	13.400
	L3	0.692	0.255	0.234	123.747	13.367

Table C1: *Full Phase 2 inverted-gold context-rot results across L0, L2, and L3.*

Condition	Method	Success	Distractor Selection Failure	Evidence Localization Failure	Other Failures	Total
Phase 2 inverted-gold	B RAG	8	48	4	0	60
	C full	**22**	**30**	6	2	60
Phase 3 semantic	B RAG	45	7	6	2	60
	C full	54	3	**3**	0	60
	C no-tools	**55**	**2**	**3**	0	60

C no-followup	48	3	7	2	60

Table C2: *Condensed failure taxonomy counts under L3 conditions*